Stress and Its Contributors, by John Adesola and Ruth Kayinda

Introduction

Stress plays a significant role in modern society, affecting individuals' physical health, mental well being and overall productivity. By understanding the factors that contribute to stress, we are able to improve the quality of life and promote healthier lifestyles. In this study, we analyse the the Stress Level Prediction dataset, containing demographic and lifestyle variables, including sleep habits, physical activity, screen time, and work-related factors, to better understand their relationship with stress levels. We aim to both predict stress levels, and identify key relationships between stress and its contributing factors, by making use of various statistical techniques including multiple linear regression, Chi-square tests, and multivariate analysis of variance (MANOVA).

Description of The Data

For this study, we will analyze the Stress Level Prediction dataset obtained from Kaggle. It includes demographic and lifestyle variables such as:

- Age: Age of the individual (in years)
- Gender: Gender of the individual (categorical variable)
- Occupation: Type of occupation or employment category
- Sleep duration: Average number of hours of sleep per day
- Sleep quality: Rating or category describing quality of sleep
- Physical activity: Level or frequency of physical activity
- Screen time: Average amount of time spent on electronic devices per day.

The dataset consists of multiple observations, where each row represents an individual and each column corresponds to a specific variable describing that individual's characteristics or behavior. The dataset includes both continuous variables (age, sleep duration, screen time) and categorical variables (gender, occupation). The response variable in our analysis will be stress level, which is treated as a quantitative variable in our analysis, and which we aim to model and predict using available predictors.

Multiple Linear Regression Model

Since the response variable, Stress_Detection, can be expressed in the form of numerical values (for example, Low = 1, Medium = 2, High = 3), multiple linear regression is one of the most effective means to interpret this data. We are fitting the model: Stress_Level= 0+ 1(Age)+ 2(Gender)+ 3(Sleep Duration)+ 4(Sleep Quality) + 5(Physical Activity)+ 6(Screen Time)+ 7(Caffeine Intake)+ 8(Work Hours)+ 9(Alcohol Intake)+

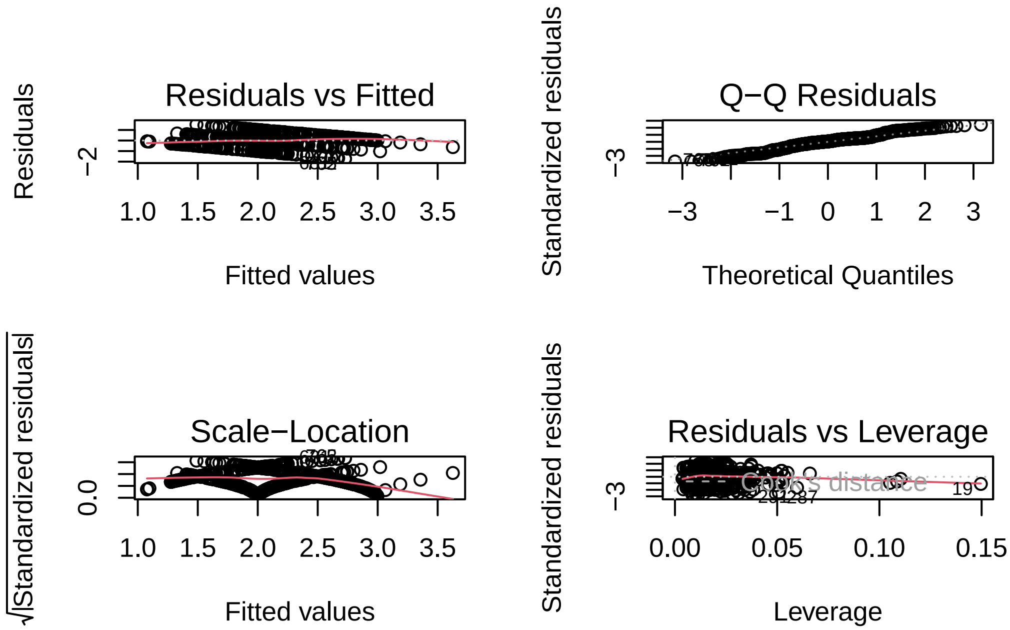
```r
#1. MULTIPLE REGRESSION : Model stress as numeric, interpret coefficients

stress <- read.csv("stress_detection_data.csv")
stress$Stress_Level <- as.numeric(factor(stress$Stress_Detection,
                                          levels = c("Low","Medium","High")))
stress$Gender <- as.factor(stress$Gender)

set.seed(123)
idx <- sample(1:nrow(stress), 0.8 * nrow(stress))
train <- stress[idx, ]; test <- stress[-idx, ]

lm_model <- lm(Stress_Level ~ Age + Gender + Sleep_Duration + Sleep_Quality +
                 Physical_Activity + Screen_Time + Caffeine_Intake +
                 Work_Hours + Alcohol_Intake, data = train)
summary(lm_model)
```

```
Call:
lm(formula = Stress_Level ~ Age + Gender + Sleep_Duration + Sleep_Quality +
    Physical_Activity + Screen_Time + Caffeine_Intake + Work_Hours +
    Alcohol_Intake, data = train)

Residuals:
    Min      1Q  Median      3Q     Max
-1.7282 -0.3343  0.0332  0.3251  1.5106
```

```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         0.6327685  0.3408024   1.857 0.063837 .  
Age                 0.0041060  0.0036191   1.135 0.257019    
GenderMale         -0.0563922  0.0545999  -1.033 0.302096    
Sleep_Duration     -0.1645534  0.0397898  -4.136 4.04e-05 ***
Sleep_Quality       0.0003082  0.0483029   0.006 0.994912    
Physical_Activity   0.0527377  0.0529506   0.996 0.319657    
Screen_Time         0.1772921  0.0531287   3.337 0.000898 ***
Caffeine_Intake     0.1427144  0.0489725   2.914 0.003697 ** 
Work_Hours          0.1405645  0.0272197   5.164 3.28e-07 ***
Alcohol_Intake      0.1557052  0.0523985   2.972 0.003080 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6203 on 608 degrees of freedom
Multiple R-squared:   0.34, Adjusted R-squared:  0.3302 
F-statistic:  34.8 on 9 and 608 DF,  p-value: < 2.2e-16
```

The total model generated is statistically significant ($F(9, 608) = 34.8$, $p < 2.2 \times 10^{\ 1}$), and one predictor is associated with an individual's stress levels. The model indicates moderate fit. sleep duration significantly negatively relates to stress ($\ = -0.165$, $p < 0.001$) indicating, more sleep has less stress. Interestingly, however, screen time and caffeine intake, hours of work, and alcohol consumption are positively associated with stress, thus greater levels of stress are correlated with higher levels of these factors. Meanwhile, age, gender, sleep quality, and physical activity were not statistically significant, indicating that they are not significant predictors of stress, even when control for all other variables.

```r
#plots
par(mfrow = c(2, 2))
plot(lm_model)
```



The plots demonstrate that certain assumptions of linear regression are not adequately met. The residual vs fitted plot indicate that the model is not linear. The plot of coordinates on the scale shows the non-constant variance of the residuals. The Q-Q plot shows a slight deviation from the normal distribution at the tails. According to the residuals vs leverage

plot, little effect on any two of those things can be distinguished from other sources, or moderate outliers.

Training and Testing the Model

We evaluate our Multiple Linear Regression's model performance by training and testing the model. This consists of splitting the dataset into a Training set, for model building, and a testing set for Model evaluation on unseen data. We want to ensure that our model is not just memorizing the data, but can as well generalize into new observations

```r
#import dataset
data <- read.csv("stress_detection_data.csv")

#set some variables to categorical
data$Gender <- as.factor(data$Gender)
data$Occupation <- as.factor(data$Occupation)
data$Smoking_Habit <- as.factor(data$Smoking_Habit)
data$Marital_Status <- as.factor(data$Marital_Status)
data$Exercise_Type <- as.factor(data$Exercise_Type)
```

```r
#Stress level as numeric

data$Stress_Level <- as.numeric(factor(data$Stress_Detection,
                                       levels = c("Low", "Medium", "High")))
#seed value
set.seed(123)

#80/20 train/test split
train_index <- sample(1:nrow(data), 0.8*nrow(data))

train <- data[train_index, ]
test <- data[-train_index, ]

# model

model <- lm(Stress_Level ~ Age + Gender + Sleep_Duration +
                 Sleep_Quality + Physical_Activity + Screen_Time +
                 Caffeine_Intake + Work_Hours + Alcohol_Intake,
                 data = train)

#prediction

pred   <- predict(model, newdata = test)

# model evaluations


mse  <- mean((test$Stress_Level - pred)^2)

mse
```

```
[1] 0.3745672
```

We look at the Mean Squared Error (MSE) to measure the average squared difference between actual and predicted values. Looking at the MSE value of 0.3745672, this indicates to us that the predicted stress levels are reasonably close to the observed values. This suggests reasonable performance of our model on real-world data, and good predictive capability, with some variability.

Chi-Square Test

We now aim to determine whether there is a significant relationship between the variables Stress Level and Marital Activity, as well as Stress Level and Physical Activity.

Hypotheses:

Stress Level vs Martial Status

H : Stress level and marital status are independent H : They are associated

Stress Level vs Physical Activity

H : Stress level and Physical activity are independent H : They are associated

Stress level vs Marital activity

```r
# make sure the variables are categorial

data$Stress_Level    <- as.factor(data$Stress_Level)
data$Marital_Status  <- as.factor(data$Marital_Status)

# Contingency table
table_marital <- table(data$Marital_Status, data$Stress_Level)

# Chi-square test
chi_marital <- chisq.test(table_marital)

# Output
table_marital
```

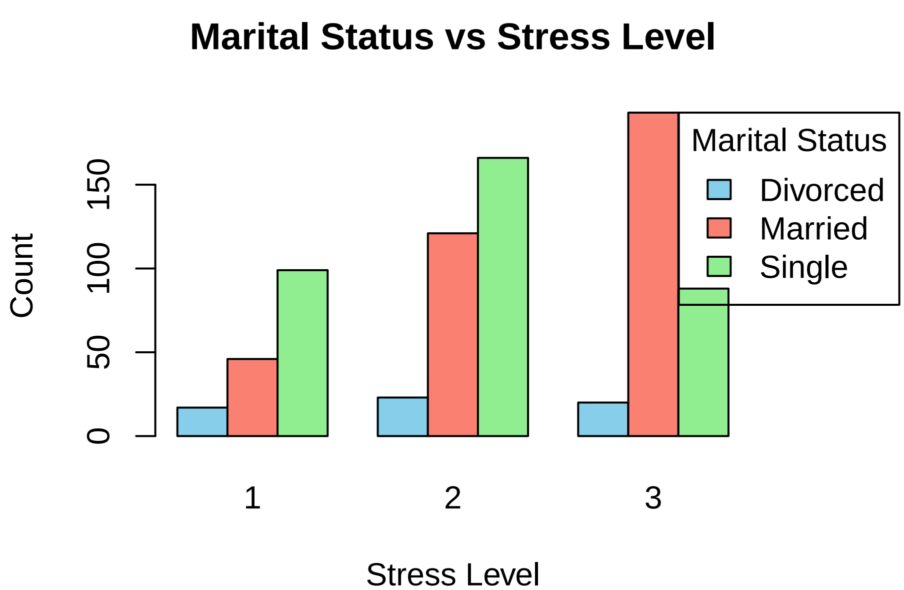
```
            1   2   3
  Divorced  17  23  20
  Married   46 121 193
  Single    99 166  88
```

```r
chi_marital
```

```

	Pearson's Chi-squared test

data:  table_marital
X-squared = 67.74, df = 4, p-value = 6.805e-14
```

From our Chi square test, we have a p-value of 6.805e-14, which is far less than our level of significance of 0.05. Therefore we reject the null hypothesis and conclude that marital status and stress level have a significant relationship. This suggests to us that marital status may play a role in inluencing stress.

```
# bar plot
par(mar = c(5,4,4,8))

barplot(table_marital,
        beside = TRUE,
        col = c("skyblue", "salmon", "lightgreen"),
        legend.text = rownames(table_marital),
        args.legend = list(title = "Marital Status",
                           x = "topright",
                           inset = c(-0.25, 0)),
        main = "Marital Status vs Stress Level",
        xlab = "Stress Level",
        ylab = "Count")
```



From our bar plot, we see some noticeable differences in the distribution of stress levels across marital groups. This supports our findings from the Chi-square test.

Stress Level vs Physical Activity

```r
# Convert Physical Activity into categories
data$PA_Level <- cut(data$Physical_Activity,
                     breaks = c(0, 2, 3, Inf),
                     labels = c("Low", "Moderate", "High"),
                     right = TRUE)

# Convert to factor
data$PA_Level <- as.factor(data$PA_Level)

# Contingency table
table_pa <- table(data$PA_Level, data$Stress_Level)

# Chi-square test
chi_pa <- chisq.test(table_pa)

# Output
table_pa
```

```
           1   2   3
Low       73  88  36
Moderate  80 192  95
High       9  30 170
```

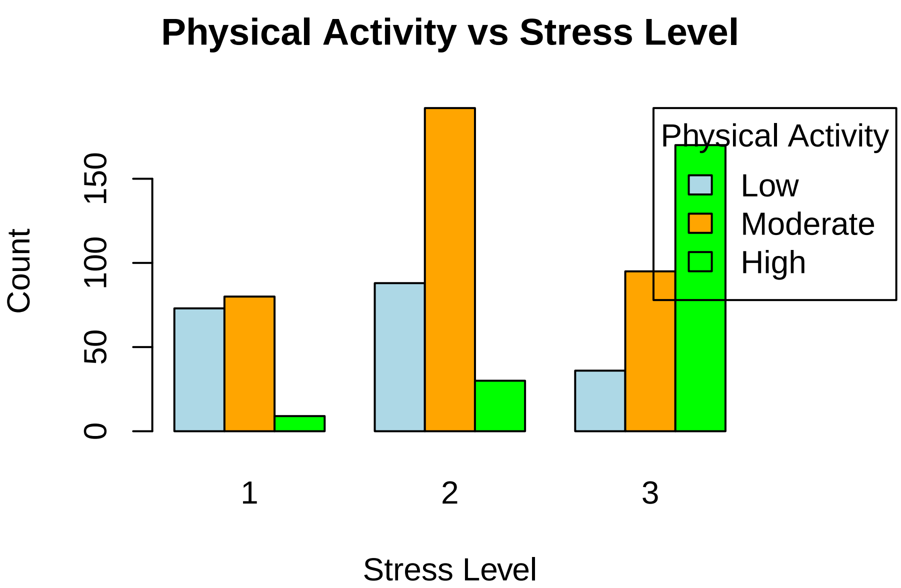
```r
chi_pa
```

```
    Pearson's Chi-squared test

data:  table_pa
X-squared = 235.52, df = 4, p-value < 2.2e-16
```

Similarly, in the Chi-square test for association between Stress Level and Physical Activity, since we have p-value $< 2.2\text{e-}16 < 0.05$, we reject the null hypothesis and conclude that physical activity level and stress have a significant relationship. The distribution of stress levels differs signifcantly across groups, as we see that individuals with high activity have notably higher levels of stress.

```r
#bar plot
par(mar = c(5,4,4,8))
```

```r
barplot(table_pa,
        beside = TRUE,
        col = c("lightblue", "orange", "green"),
        legend.text = rownames(table_pa),
        args.legend = list(title = "Physical Activity",
                           x = "topright",
                           inset = c(-0.25, 0)),
        main = "Physical Activity vs Stress Level",
        xlab = "Stress Level",
        ylab = "Count")
```



The bar plot supports our findings in the Chi-square test, again reinforcing the differences in stress level distributions across different activity groups. We see a particularly large concentration of high stress levels among individuals who are highly active. This is an interesting observation, as most would assume lower stress levels for highly active individuals. This suggests underlying factors such as workload and lifestyle demands within our dataset.

Basic Hypothesis Test (ANOVA)

We use Analysis of Variance to determine whether there are significant differences in the mean values across the Stress Level and Physical Activity groups.

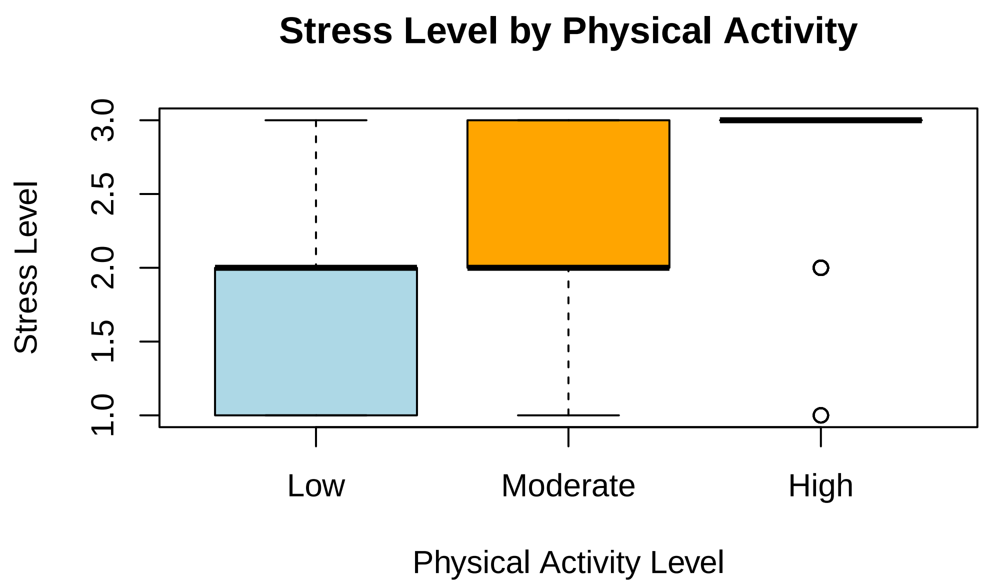
H : Mean stress is the same across all physical activity levels **H :** At least one group has a different mean stress

```r
#Physical Activity groups as before

data$PA_Level <- cut(data$Physical_Activity,
                     breaks = c(0, 2, 3, Inf),
                     labels = c("Low", "Moderate", "High"))

data$PA_Level <- as.factor(data$PA_Level)


# runnning ANOVA model

data$Stress_Level <- as.numeric(factor(data$Stress_Detection,
                                       levels = c("Low", "Medium", "High")))


anova_model <- aov(Stress_Level ~ PA_Level, data = data)
summary(anova_model)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
PA_Level      2  106.6   53.30   123.8 <2e-16 ***
Residuals   770  331.4    0.43
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From our anova test, since our p-value $< 2\text{e-}16 < 0.05$, we reject the null hypothesis and conclude that the mean stress levels differ significantly across physical activity levels. These results are consistent with our Chi-Square analysis, which also indicate a significant relationship between physical activity and stress levels.

```r
# boxplot

boxplot(Stress_Level ~ PA_Level,
        data = data,
        col = c("lightblue", "orange", "lightgreen"),
        main = "Stress Level by Physical Activity",
        xlab = "Physical Activity Level",
        ylab = "Stress Level")
```



Our boxplot indicates differences in stress level disstributions across the different phycial activity groups. This supports our ANOVA findings. Similarly to our bar plot, the box plot shows an increase in stress levels as physical activity increases, with the high activity group showing the highest median level of stress.

Hotellings T^2 Test

Hotelling's T^2 test comparison of males and females at the same time was conducted at different time points, with the aim to compare males and females across a number of variables, such as sleep duration, screen time and hours of work.

It defines the overall lifestyle profile as gender-specific sex differences.

```
library(mvtnorm)
library(ICS)
library(ICSNP)
males <- stress[stress$Gender == "Male", c("Sleep_Duration","Screen_Time","Work_Hours")]
females <- stress[stress$Gender == "Female", c("Sleep_Duration","Screen_Time","Work_Hours")]
HotellingsT2(males, females)
```

```

	Hotelling's two sample T2-test

data:  males and females
T.2 = 30.31, df1 = 3, df2 = 769, p-value < 2.2e-16
```

```
alternative hypothesis: true location difference is not equal to c(0,0,0)
```

The findings indicate that there is a significant males-female difference in the overall level ($T^2 = 30.31$, $p < 2.2 \times 10^{1}$).

MANOVA

We test through the MANOVA to check whether the stress levels (Low, Medium, High) groups were significantly different based on the lifestyle factors like duration of sleep, screen time, hours worked, physical activity and lifestyle factors.

```r
#MANOVA : Test if stress groups differ on multiple lifestyle vars

manova_model <- manova(cbind(Sleep_Duration, Screen_Time, Work_Hours, Physical_Activity)
                       ~ Stress_Detection, data = stress)
summary(manova_model, test = "Wilks")
```

```
                  Df   Wilks approx F num Df den Df    Pr(>F)    
Stress_Detection   2 0.65527   45.128      8   1534 < 2.2e-16 ***
Residuals        770                                             
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

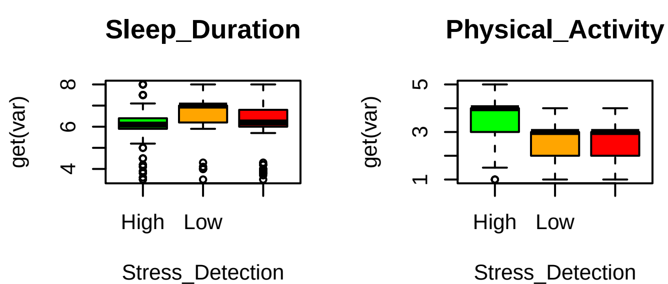
```r
summary.aov(manova_model)  # Follow-up univariate ANOVAs
```

```
 Response Sleep_Duration :
                  Df Sum Sq Mean Sq F value   Pr(>F)    
Stress_Detection   2  27.07 13.5342  26.833 5.43e-12 ***
Residuals        770 388.38  0.5044                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Response Screen_Time :
                  Df Sum Sq Mean Sq F value    Pr(>F)    
Stress_Detection   2 131.61  65.806  134.03 < 2.2e-16 ***
Residuals        770 378.04   0.491                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
 Response Work_Hours :
                  Df Sum Sq Mean Sq F value    Pr(>F)
Stress_Detection   2  75.83  37.916  36.566 6.745e-16 ***
Residuals        770 798.42   1.037
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Response Physical_Activity :
                  Df Sum Sq Mean Sq F value    Pr(>F)
Stress_Detection   2 105.87  52.937  105.93 < 2.2e-16 ***
Residuals        770 384.80   0.500
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Analysis (Wilks' Lambda) suggests that there is a significant overall difference ($p < 2.2 \times 10^{1}$) between stress groups, indicating that at least one of these lifestyle factors varies along stress levels. The variance in ANOVA tests ($p < 0.001$) suggest that the sleep duration, screen time, work hours, and physical activity significantly vary between the stress groups. However, we see that the differences in stress level are closely proportional to the types of lifestyle behaviours.

```
par(mfrow = c(2, 3))
for (var in c("Sleep_Duration", "Physical_Activity" )) {
  boxplot(get(var) ~ Stress_Detection, data = stress, main = var, col = c("green", "orange",
}
```



The boxplots also reveal people under high stress are sleeping much shorter and on average, less active overall than those under low stress, who generally sleep more and are more active. This agrees with the MANOVA conclusion that poor lifestyle choices were related to poor stress.

Conclusion

In this study we took a look at the factors contributing to stress, using the Stress Level Prediction Dataset. Through use of Multiple Linear regression, we observed that variables including sleep duration, screen time, caffeine intake, work hours, and alcohol intake play a significant role in predicting stress. Our findings highlight the important role that lifestyle behaviors have in influencing stress. Overall, we see that daily habits such as sleep patters, work demands, and activity levels are strongly linked to stress levels. Improving these factors may aid in reducing stress and enhance overall well-being.